

Climate Change Mitigation

Group 10

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Project

- Estimate the year by which we can have a circular economy for the Nickel required for EV.
- Circular economy for the Nickel required by EV, with a circular economy defined as 90% of the Nickel used for batteries comes from spent batteries.

Questions:

1. How much Ni is needed per EV?
2. Annual Ni demand? (US + global)
3. Determine the percentage of recycled Ni in the supply chain as a function of year.
4. When is circular Ni economy achieved?

Q1 : Nickel requirement per EV

Voltage 3.5 V

NMC811 $\rightarrow$ 0.8 mol Ni per mol e^- .

300 miles of range

58.693 atomic mass units for Ni

Efficiency – 0.276kwh/mi

= Total 39kg of Ni

1. Assume $V \sim 3.5$ volts Range 250 miles – 300 miles
I have to pick (justification)
Boston $\rightarrow$ New York
Might have other answer.
(Avg. Car range)

NMC811 = 0.8
Faraday - ~~2~~ $F = 2.8 \text{ A} \cdot \text{h} \cdot \text{mol}^{-1} e^-$
Efficiency = 33.7 kWh

One car = 15000 lbs
how much nickel do you need
how many cars (X) 300m 200m 40m
(Avg. home hold)
1.7 ~ 2 Cars.
340 millions
3
10 years
15 million.

Average efficiency of an electric car.
(m/kwh) (Efficiency)
gov website. 0.276kwh/mi (172)
3 mi/
0.27 $\frac{\text{kWh}}{\text{mi}} \times 300 \text{ mi} = 81 \text{ kWh}$ 276 3.6 m/kwh
 $Ah = \frac{81,000 \text{ Wh}}{3.5 \text{ V}} = 23,143 \text{ Ah}$
 $n_e = \frac{Ah}{F} = \frac{23,143}{2.8} = 826.54 \text{ me}^-$
 $m_{Ni} = 661 \times 58.69 = 38,807.5$
 $\approx 39 \text{ kg of Ni}$ $n_{Ni} = 0.8 \times 826.54 \approx 661$

Q2: Kg of Ni to switch to EV (USA estimate)

Reason for not getting EV	% of population (350 million people)
Access to public transit	10%
Not affordable	20%
Already have EVs	5%

In summary: 35% won't get EVs while 65% of people will be in the market for one. $350 \text{ million} \times 65\% = 227,500,000$ people in the market for an EV. Average of 4 people per household and 1 car per household. $227,500,000 / 4 = 56,875,000$ EVs.

However, on average EVs have a lifespan of 15 years. $56,875,000 \text{ EV} \times (1\text{EV}/15 \text{ years}) = 3,791,667$ EVs need to be manufactured per year.

$3,791,667 \text{ EVs} \times (39 \text{ kg Ni}/\text{EV}) = 147,875,013 \text{ kg Ni}$

Q2: Kg of Ni to switch to EVs (Global Estimate)

Total population: 8 billion. 25% of people live in cities with good public transit or just don't need a personal vehicle in their daily lives (higher % than the US because Europe and Asia are less car dependent on average). 20% of people can't afford an EV. 3% of people already have EVs (less than before because EVs are less common outside of the US).

In summary: 48% won't get EVs while 52% of people will be in the market for one. $8 \text{ billion} * 52\% = 4.16e9$ people in the market for an EV. Average of 4 people per household and 1 car per household. $4.16e9 / 4 = 1.04e9$ EVs.

However, on average EVs have a lifespan of 15 years. $1.04e9 \text{ EV} * (1\text{EV}/15 \text{ years}) = 69,333,333$ EVs need to be manufactured per year.

$69,333,333 \text{ EVs} * (39 \text{ kg Ni/EV}) = 2.7e9 \text{ kg Ni per year}$

Q2: Comparing demand to global production from mines

Global production of nickel from mines: 3.6 million tonnes = 3.6×10^9 kg Ni

This is a higher value than the global demand (2.7×10^9 kg)

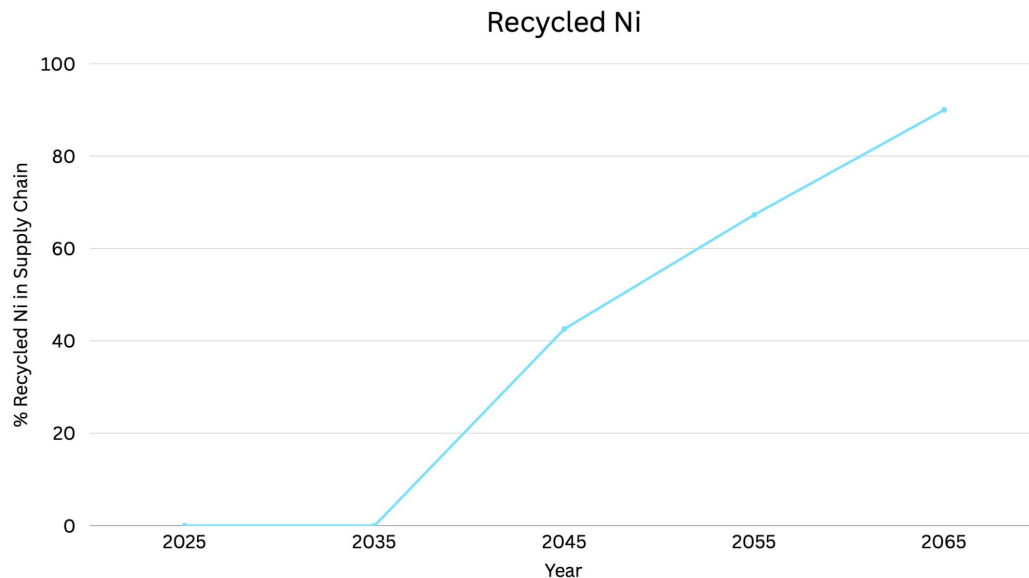
So the number of mines that exist currently is enough to satisfy this demand for EV manufacturing

Q3: Is 100% EV by 2050 Realistic?

These estimates are not realistic because of many factors. Firstly, not every person who owns a car that runs on gasoline can afford to pay for an electric car. Secondly, there might not be enough mines in the world to mine enough Nickel for 100% of cars to be EVs by the year 2050. The estimate of 10% of new cars globally will be fully electric EVs starting this year seems somewhat realistic since not everyone would be willing to buy a fully electric car so only about 10% seems fair. This is because electric cars are becoming more popular.

Q3: Percentage of Recycled Ni

Now assume that we can recover 90% of the Nickel from a battery. Determine (and plot) the percentage of recycled Ni in the supply chain as a function of year. State all assumptions. Assumption: The lifespan of a battery is 12 years.



Q4: Commentary

Some environmental and climate tradeoffs of switching from internal combustion engines to EV are that electric cars might be better for the environment since they do not release greenhouse gases but making them already causes more damage due their production phase emitting more greenhouse gases.

Q4: Recommendations

One recommendation on how to facilitate the transition from gasoline-fueled cars to fully electric EVs is to look into how to minimize the greenhouse gas emissions that are produced through the production of the EVs.

Another recommendation would be to add more charging stations for the EVs as well as make these charging stations more accessible.